

Tutorial Problem Set XII

MATA30 – TUT0003

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In this last tutorial we discussed integration techniques and some further applications of the fundamental theorem of calculus. Best of luck with all your future endeavours, both mathematically and otherwise!

1 Integration techniques

Problem 1. Compute

$$\int_2^{e+1} \frac{x}{(x-1)^2} dx.$$

Problem 2. Compute

$$\int \frac{1}{x^2 + x + 1} dx.$$

Problem 3. Compute

$$\int_{-1}^1 \sqrt{4-x^2} dx.$$

Problem 4. Compute

$$\int \cos(\log(x)) dx.$$

2 Fundamental theorem of calculus

Problem 5. Let $F(x) = \int_0^1 \frac{\sin(tx)}{t} dt$.

(a) Compute $F'(x)$.

(b) Compare with

$$\int_0^1 \frac{\partial}{\partial x} \left(\frac{\sin(tx)}{t} \right) dt.$$

(c) Can you find suitable assumptions on $f : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}$ such that

$$\frac{d}{dx} \left(\int_0^1 f(x, t) dt \right) = \int_0^1 \frac{\partial f}{\partial x}(x, t) dt?$$

Problem 6. In this series of exercises we will prove the following inequality.

Theorem 1 (Grönwall's inequality). *Suppose that $U : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function and that $K : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function such that $U'(x) \leq K(x) \cdot U(x)$ for all $x \geq 0$. Then*

$$U(x) \leq U(0) \cdot e^{\int_0^x K(x)}.$$

for all $x \geq 0$.

- (a) Compute the derivative of the function $F : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(x) := U(x) \cdot e^{-\int_0^x K(x)}.$$

- (b) Show that F is decreasing and deduce Grönwall's inequality.
- (c) (Bonus) Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is *Lipschitz*, i.e. that there exists a constant $K \in \mathbb{R}$ such that $|f(x_1) - f(x_2)| \leq K|x_1 - x_2|$ for all $x_1, x_2 \in \mathbb{R}$. Consider the initial value problem

$$\begin{cases} y'(x) = f(x) \\ y(0) = a \end{cases} \quad (\dagger)$$

for some $a \in \mathbb{R}$. Prove that this initial value problem has at most one solution $y : \mathbb{R} \rightarrow \mathbb{R}$ using Grönwall's inequality.